Conclusion

Conclusion I

`template_autoresearch_project` shows how a tractable AutoResearch task can be represented as reproducible template infrastructure. The `small MNIST neural-network classification` experiment is small by design, but it exercises the method surfaces that matter for a public default: a declared research program, local input-data provenance, bounded candidate families (`MLP, nearest-centroid, softmax regression, tiny patch-attention`), objective scoring, budget and cost ledgers, evidence-linked claims, generated figures, benchmark grading, manuscript-variable hydration, loop-settlement ledgers, figure-quality checks, validation, local integrity attestation (`passed`), and human review gates.

Conclusion II

The broader field is converging across five related trends. Autoresearch systems seek end-to-end automation of ideation, experiment execution, writing, and review. Autoformalization pushes informal claims toward machine-checkable representations. ML-for-ML systems use search, evaluation, and evolutionary code loops to improve algorithms. Agentic science composes retrieval, planning, experimentation, and communication into higher-autonomy workflows. Structured knowledge synthesis supplies the substrate that all of those systems need: source-linked artifacts, knowledge graphs, tensors, vector spaces, and uncertainty-aware memories that can be navigated without losing provenance.

Conclusion III

This exemplar makes a deliberately narrower contribution. It does not mine the live literature, construct a Conceptual Nexus Model, search for Lean proofs, mutate arbitrary code, coordinate external agents, or approve its own paper. Instead it demonstrates governance infrastructure for automated research workflows: every important claim is hydrated from run artifacts, every generated figure is registry-bound and locally checked for source and pixel integrity, every budget and candidate decision is recorded, and review state remains outside generated self-approval. The security layer adds local inventory and checksum evidence while keeping external signing and production deployment claims out of scope.

Conclusion IV

The durable product is therefore not only the `MNIST` metric. It is a reproducible research process whose data, claims, captions, figures, review gates, and manuscript variables can be reviewed, rerun, and extended. As more ambitious automated-science systems mature, this kind of offline, deterministic, evidence-governed baseline remains useful because it preserves the part of scientific automation that should not be optional: inspectable provenance, explicit limits, and human-governed publication judgment.